

COLLEGE OF POPULATION STUDIES • CHULALONGKORN UNIVERSITY

Demography in the Data and AI Era

Session 1 • Introduction to Fundamental Demographic Analysis

Course 5140602 • Fundamental Demographic Analysis (Revised 2026 Syllabus)

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3-hour session



About Your Instructor & This Course

- **Asst. Prof. Dr. Ruttiya Bhula-or, College of Population Studies, Chulalongkorn University**
- Research interests: population and labour, ageing, demographic methods, and applied data analysis
- This course trains you to think like a demographer: to count, measure, and explain population change — and to critically evaluate the data behind every number you use
- A Master's-level, methods-first course: expect worked calculations, real datasets, and applied research skills throughout the semester



Getting to Know Each Other

- In pairs, take 3 minutes each to share:
 - 1. Your name, programme, and one country/region whose population trends fascinate you
 - 2. One number about population (any number!) that has stuck with you — and why
 - 3. One question you hope this course will answer
- We will hear 4–5 pairs report back to the whole class



8 minutes

Total activity time



By the End of This Session, You Will Be Able To...



Define demography

Explain the discipline's scope and distinguish formal demography from population studies



Use the basic toolkit

Correctly compute and interpret counts, proportions, ratios, and rates



Explain the demographic transition

Describe the stages of the transition model and critique its universality



Situate demography in the data/AI era

Identify how traditional and emerging data sources reshape demographic evidence



Apply academic integrity norms

Use generative AI tools responsibly and transparently in coursework



Today's 3-Hour Structure

- Welcome, icebreaker & learning objectives
- Block 1 — What is demography, and why study it?
- Block 2 — The demographer's core toolkit (counts, ratios, rates)
- Break
- Block 3 — Demographic transition theory
- Block 4 — Demography in the data/AI era + course logistics
- Wrap-up, Q&A, practice exercises

**3:35–4:00 relative to a 4:00 total (approx.); timings are indicative and may flex in class.*



Syllabus at a Glance: 15 Sessions

Segment	Sessions	Focus
Foundations	1–2	Introduction; sources of demographic data
Mortality & morbidity	3–4	Health transitions; life tables
Fertility & migration	5–6	Fertility measurement; migration analysis
Population structure	7	Age–sex structure, population pyramids
Midterm examination	8	Covers Sessions 1–7
Applied classes	9–13	Household/union analysis; ageing measurement; AI & digital-data methods (11–12); integration (13)
Synthesis & review	14	Applied research paper workshop
Final examination	15	Comprehensive, covers full semester



How Your Grade Is Assessed

Assessment weights for 5140602 Fundamental Demographic Analysis (2026 revised syllabus):

- Midterm examination — 30%
- Final examination — 40%
- Computational labs / Assignment — 20%
- Participation — 10%

Computational labs and the applied research paper are your main venues for hands-on, disclosed use of demographic software and AI tools.



Expectations & How We'll Work Together

- Come having skimmed assigned readings — class time is for application, not first exposure
- Bring a laptop with spreadsheet software for in-class calculations and computational labs
- Office hours: by appointment — email to schedule, response within 2 business days
- All coursework must disclose any generative-AI assistance used, per the academic-integrity policy discussed today
- **Questions about grading, accommodations, or wellbeing are always welcome — please reach out early, not after a deadline has passed**

BLOCK 1

What Is Demography, and Why Study It?

Etymology, definitions, the boundary between formal demography and population studies, and why this evidence base matters for policy and society.



The Etymology of "Demography"

- "Demography" derives from Greek: demos (δῆμος, "the people") + graphein (γράφειν, "to write/describe")
- Literally: "the writing/description of the people"
- First coined in this modern sense by Achille Guillard (1855), a Belgian statistician, in *Éléments de statistique humaine ou démographie comparée*
- From the outset, demography combined description (counting, measuring) with comparison across populations and over time — a dual identity it retains today



Two Faces of the Discipline

Formal (Pure) Demography

- Mathematical/statistical study of population size, structure, and change
- Focuses on the three direct drivers: fertility, mortality, migration
- Builds models: life tables, stable population theory, population projections
- Answers: "What are the mechanics of population change?"

Population Studies (Social Demography)

- Draws on economics, sociology, epidemiology, geography to explain causes and consequences of demographic patterns
- Asks why fertility declines, why migration happens, how ageing reshapes welfare systems
- Answers: "Why does population change happen, and what does it mean for society?"



Computational Questions (6–9)

- 6. A country had 480,000 live births and a mid-year population of 60,000,000. Calculate the crude birth rate per 1,000.
- 7. If a population has 8,200,000 people aged 65+ and 42,000,000 people aged 15–64, calculate the old-age dependency ratio.
- 8. A district recorded 1,050 male births and 1,000 female births in one year. Calculate the sex ratio at birth (males per 100 females).
- 9. If the crude death rate is 8.5 per 1,000 and the mid-year population is 20,000,000, how many deaths occurred that year?



The Demographic Perspective: Core Domains



Fertility

Childbearing patterns, family formation, reproductive health and policy



Mortality

Death, survival, life expectancy, and health transitions



Migration

Internal and international movement, displacement, urbanization



Population structure

Age-sex composition, dependency, population momentum



Population change

Growth, decline, distribution over time and space



Policy & society

Links to health, economic, environmental, and social policy

Why Study Demography?

For example: Thailand

20%

of Thailand's population is aged 60+ as of 2024 — Thailand is now a "complete aged society"

2035

projected year Thailand becomes a "super-aged society" ($\geq 20\%$ aged 65+)

58:42

female-to-male ratio among Thai elderly — with major implications for care policy

EXAMPLE 1

Thailand's Aged Society, 2024



- Thailand's share of population aged 60+ rose from 6.8% (1994) to 20% (2024) — one of the fastest ageing trajectories in Asia
- Of 14 million Thai seniors, only 4.76 million remain economically active; 63.7% of working seniors are self-employed
- These are not abstract numbers: they drive pension reform, long-term care policy, labour-market design, and health-system planning
- This is exactly the kind of question formal demography (measuring the age structure) and population studies (explaining causes, designing policy) must answer together

LET'S DISCUSS

Formal Demography or Population Studies?



- A national statistical office projects that the working-age population will shrink 15% by 2040 using a cohort-component model. Which branch is this?
- A sociologist studies why young Thai adults are delaying marriage and childbearing. Which branch is this — and could it also use formal-demographic tools?
- Can you think of a real research question that requires BOTH branches working together?



Conceptual Questions (1–5)

- 1. In your own words, define demography and explain the Greek roots of the term.
- 2. Distinguish formal demography from population studies using an example research question for each.
- 3. Explain the difference between a proportion and a ratio, giving one demographic example of each.
- 4. Why is a "rate" considered a measure of the pace of events, and what does the denominator represent?
- 5. Name two events in a person's life course that formal demography treats as core measurable events.

BLOCK 2

The Demographer's Core Toolkit

Four foundational measurement concepts — count, proportion, ratio, and rate — that underlie every indicator you will compute this semester.



Four Building Blocks of Measurement

Concept	Definition	Example
Count	A simple enumeration — the raw number of events or persons	Number of live births in Thailand in 2024
Proportion	A part relative to a whole that includes that part (numerator $\subset$ denominator)	Proportion of the population aged 60+ (0.20)
Ratio	One quantity relative to a different, non-overlapping quantity	Sex ratio: males per 100 females
Rate	A measure of the pace of an event per population "at risk," usually over time	Crude birth rate per 1,000 population per year



Crude Birth Rate (CBR)

$$\text{CBR} = (B / P) \times 1,000$$

- B = number of live births in a given year
- P = mid-year total population
- Result expressed as births per 1,000 population

Example: if

- B = 520,000 and
- P = 66,000,000 → $\text{CBR} = (520,000 / 66,000,000) \times 1,000 \approx 7.9$ births per 1,000



Old-Age Dependency Ratio (OADR)

$$\text{OADR} = (\text{Pop } 65+ / \text{Pop } 15-64) \times 100$$

- Numerator: population above conventional working age
- Denominator: population of conventional working age
- A ratio, not a proportion — numerator and denominator are non-overlapping groups
- Rising OADR is central to Thailand's pension and long-term-care policy debates

EXAMPLE 2

Reading Thailand's Elderly Support Ratio



- NSO surveys (conducted periodically since 1994) track the elderly population and their economic activity in detail
- 59% of Thai elderly are aged 60–69 ("young-old"); 30% are in higher age brackets
- Only 16.2% of working seniors hold formal public/private-sector jobs — most (63.7%) are self-employed, with clear implications for social-security coverage
- A single ratio (dependency ratio) hides this internal structure — good demographic analysis always asks "a ratio/rate of what, computed how?"



Some Common Indicators You Will Master This Semester



CBR / TFR

Crude birth rate; total fertility rate



CDR / IMR

Crude death rate; infant mortality rate



Life expectancy

Expected years of life from birth or age



Migration rates

In-, out-, and net-migration rates



Dependency ratios

Youth, old-age, and total dependency



Growth rate

Exponential/geometric growth, doubling time



Quick Calculation: Count, Proportion, or Rate?

Working in pairs (5 minutes), classify each statistic below as a count, proportion, ratio, or rate — then justify your answer:

- 1. "Thailand recorded 502,107 live births in 2023."
- 2. "There are 96 males for every 100 females aged 60+."
- 3. "20% of Thailand's population is aged 60 or above."
- 4. "The crude death rate was 8.2 per 1,000 population in 2023."

BREAK

15 minutes

Stretch, refill your coffee, and think of one demographic statistic you've seen in the news recently.

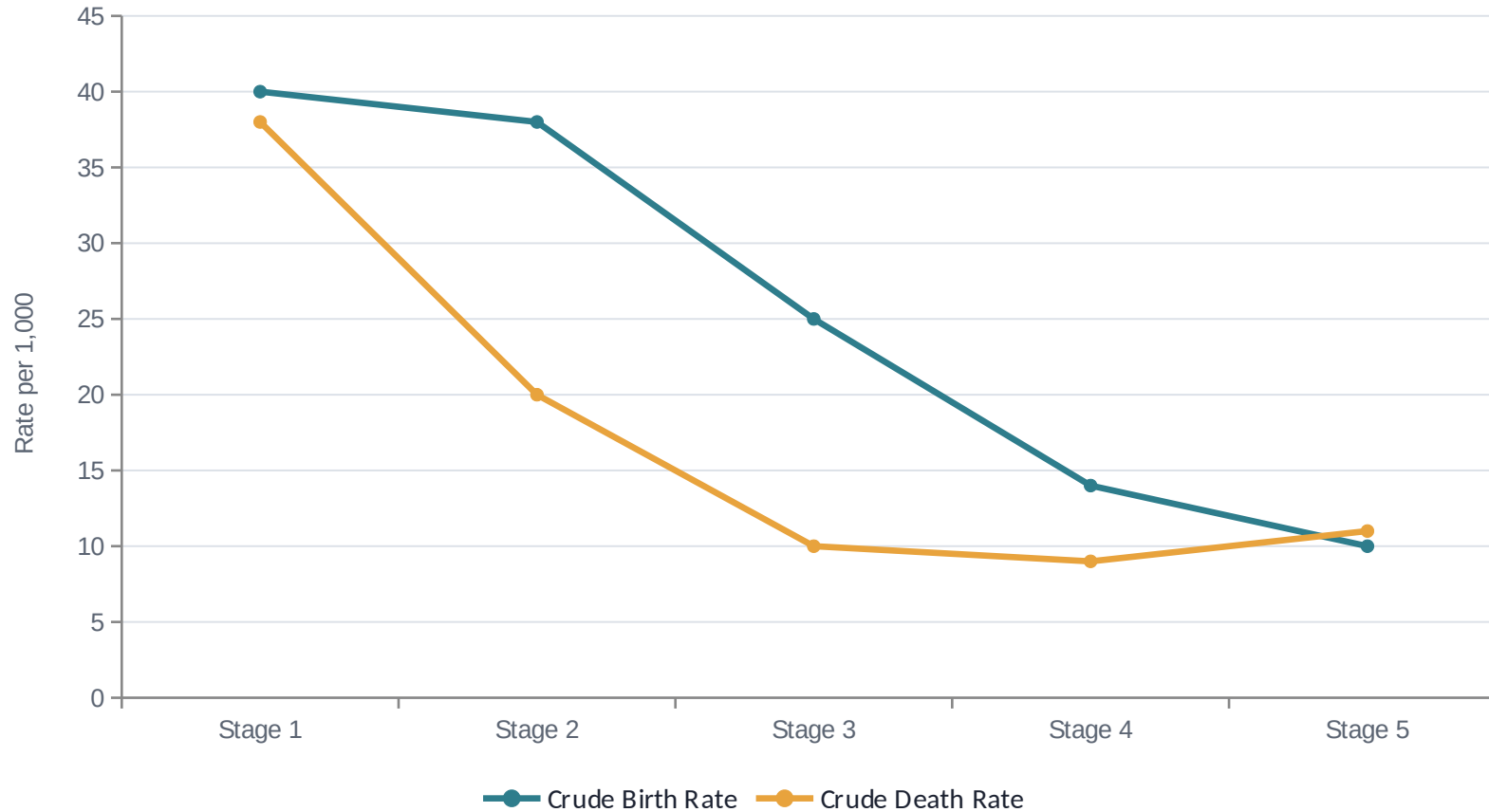
BLOCK 3

Demographic Transition Theory

The classic model of how societies move from high to low birth and death rates — its stages, evidence, critiques, and extensions.



The Classic Demographic Transition



Stage 1 — Pre-transition: high births, high deaths, slow growth

Stage 2 — Early transition: deaths fall (sanitation, medicine); births stay high → rapid growth

Stage 3 — Late transition: births begin to fall (urbanization, education, contraception)

Stage 4 — Post-transition: low births, low deaths, slow/stable growth

Stage 5 (proposed) — Sub-replacement fertility; population ageing and potential decline



Origins and Core Claims

- **First articulated by Warren Thompson (1929) and formalized by Frank Notestein (1945)**
- Describes a historical regularity observed first in Western Europe: mortality decline precedes fertility decline, producing a temporary period of rapid growth
- Kirk (1996) reviews the theory's development and shows it evolved from a loose empirical generalization into a more formal — but still contested — theory of social change
- The "engine" of the transition is debated: modernization, education (especially of women), infant-mortality decline, and ideational change have all been proposed as drivers



Strengths and Limitations of the Model

What It Gets Right

- Captures a real, broad historical pattern across most world regions
- Useful heuristic for teaching and for very-long-run population projections
- Motivated the "demographic dividend" concept used in economic-development planning

What It Misses

- Assumes a single, universal pathway — but timing, speed, and sequencing vary enormously by country
- France's fertility fell before major mortality decline — violating the classic sequence
- Says little about migration, and nothing about the "second demographic transition" (below-replacement fertility, new family forms) documented by Lesthaeghe & van de Kaa from the 1980s onward

EXAMPLE 3

Thailand's Rapid, Compressed Transition

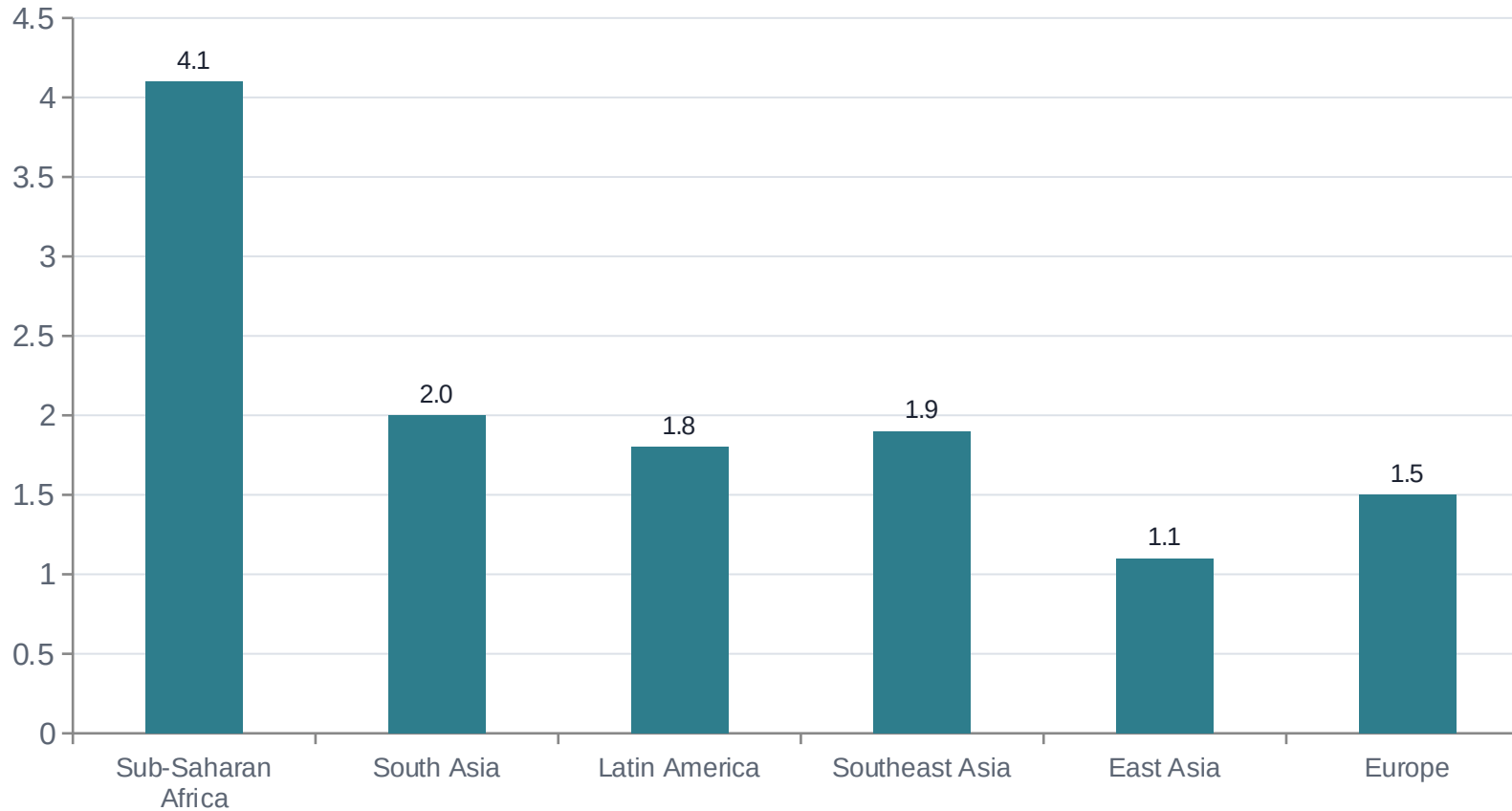


- Thailand's total fertility rate fell from roughly 6 children per woman in the 1960s to below replacement level (≈ 1.0) by the 2020s — one of the fastest fertility declines recorded globally
- Mortality decline came first (public health investment from the 1960s–70s), followed swiftly by fertility decline driven by the National Family Planning Program (from 1970) and rising female education
- The result: Thailand compressed a transition that took ~ 150 years in parts of Europe into roughly 40–50 years — and is now moving straight into Stage 5 (ageing, below-replacement fertility)
- This "telescoped" pattern is now common across East and Southeast Asia — it challenges the idea of one universal transition timeline

Source: United Nations Population Fund (2025). Thailand: World population dashboard. <https://www.unfpa.org/data/world-population/TH> ; Kirk, D. (1996). Demographic transition theory. *Population Studies*, 50(3), 361–387. <https://doi.org/10.1080/0032472031000149536>



Where Are Different Regions Today?



Illustrative total fertility rates (children per woman) by world region, mid-2020s.

Regions cluster at very different transition stages simultaneously — "the" demographic transition is really many transitions, on different clocks.

Use this to frame the discussion prompt on the next slide.

LET'S DISCUSS

Where Is Thailand — and Where Are You From?



- Based on birth and death rates, which stage of the classic model best fits Thailand today? Justify with the numbers you've seen so far.
- If you are from another country, where does it sit? Does the "second demographic transition" idea apply there?
- What would a Stage 6 look like, if population decline becomes the new normal in many countries?



Demographic Transition (10–12)

- 10. List the five stages of the classic demographic transition model and describe the birth/death rate pattern in each.
- 11. Explain one historical case (other than Thailand) where the classic transition sequence did not hold, and why.
- 12. What is the "second demographic transition," and how does it differ from the classic model's Stage 4–5?

BLOCK 4

Demography in the Data and AI Era

How the discipline's evidence base is expanding beyond censuses and surveys — and how to use generative AI responsibly in this course.



From Traditional to Augmented Data

Traditional Demographic Data

- Censuses (every 5–10 years)
- Civil registration & vital statistics (CRVS)
- Household sample surveys (e.g., DHS, LFS)
- Population registers
- High trust, official, but costly and slow to update

Emerging / Digital-Trace Data

- Mobile-phone call detail records (CDRs)
- Satellite imagery-based population mapping
- Social-media and internet-use data
- AI/ML models that fuse multiple sources
- Near-real-time, high resolution, but coverage and representativeness need careful scrutiny



Where This Theme Returns in the Course

This is a preview, not a deep dive — full methods come later in the semester:

- Session 2 — Sources of demographic data, including non-traditional/digital data (mobile CDRs, satellite imagery, social media)
- Applied Class 11-12 — Hands-on work with AI-assisted and digital-trace demographic data methods
- Applied Class 13 — Integrating traditional and non-traditional data in an applied research context
- Throughout — computational labs (15% of your grade) will build fluency with real demographic datasets and tools, including responsible use of AI assistants

EXAMPLE 4

Generative AI Is Now Widespread in Higher Education



- A 2025 systematic review of 41 studies (Jan 2021–Dec 2024) found generative AI substantially reshaping both learning and academic-integrity practice in higher education
- A related meta-analysis of 49 studies found positive average effects of GenAI use on academic achievement (effect size ≈ 0.86) and learning motivation (≈ 0.80) — but only under appropriate guidance
- Scholarship has shifted from a 2023 focus on "detection and prohibition" toward 2024–2025 emphasis on transparency, disclosure, and AI-literacy education
- Bottom line for this course: GenAI is a legitimate tool when used transparently — the norm we adopt is disclosure, not prohibition

Responsible, Disclosed Use of Generative AI in This Course



Permitted with disclosure

Brainstorming, drafting outlines, checking code syntax, or improving English clarity — cite the tool and describe your use in a short note



You own the analysis

Demographic calculations, interpretation, and argument must be your own reasoning — AI output is a starting point, never a substitute



Verify everything

AI tools can fabricate statistics and citations ("hallucinate") — every number and source you submit must be independently checked



Undisclosed use = misconduct

Submitting AI-generated work as entirely your own, without disclosure, is treated as an academic-integrity violation under university policy

EXAMPLE 5

Thailand's 2025 Census Goes Digital-First



- Thailand's National Statistical Office ran its 2025 Population and Housing Census using a "Digital First Approach" — respondents were encouraged to self-report online via a web application and government portal before enumerators followed up
- This follows earlier digital pilots in the 2020 census round, when online/mailed response rates were still low (around 5% and 4% respectively) — showing how quickly the data-collection landscape can change within one decade
- Digital-first collection previews a broader shift you will study in Session 2: official statistics increasingly blend traditional enumeration with digital tools, APIs, and administrative data



Data, AI & Course Logistics (13–14)

- 13. Name two traditional and two non-traditional/digital sources of demographic data, and one limitation of each.
- 14. Describe what "responsible, disclosed use of generative AI" means in the context of this course's academic-integrity policy.



Key Takeaways From Session 1



Demography defined

The description and analysis of population size, structure, and change — formal demography + population studies



Four core tools

Count, proportion, ratio, and rate underlie every demographic indicator you'll compute this semester



One model, many paths

The demographic transition is a useful heuristic, but real transitions vary widely in timing and sequence — Thailand's was extremely compressed



Data is expanding

Digital-trace data complements — not replaces — official statistics; more in Session 2



Integrity with disclosure

Use generative AI transparently; the analysis and interpretation must remain your own



15 sessions ahead

25% midterm, 30% final, 15% labs, 20% paper, 10% participation

QUESTIONS?

Next session: Sources of Demographic Data — from censuses to digital trace data

REFERENCES

References (1 of 2)

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Read Maltus's essay for the next class

PRACTICE EXERCISES

15 questions — conceptual & computational — to consolidate Session 1

THANK YOU

Session 2 — Sources of Demographic Data: From Censuses to Digital Trace Data

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